

$$x = \sum_{\sigma \in \Delta(k)} c_{\sigma} \sigma \quad y = \sum_{\sigma \in \Delta(k)} c_{\sigma} (1, \sigma)$$

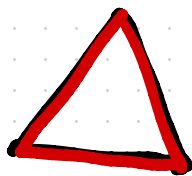
Claim: $\partial_{k+1} y = x$

Proof:

$$\begin{aligned} \partial_{k+1} \sum_{\sigma \in \Delta(k)} c_{\sigma} (1, \sigma) &= \sum_{\sigma \in \Delta(k)} c_{\sigma} \sigma + 0 \\ &= x \end{aligned}$$

because $x \in \ker \partial_k$

Example: $\Delta = 2^{[n]}/[n]$.



$$0 \xrightarrow{\partial_{n-1}} C_{n-2} \xrightarrow{\partial_{n-2}} C_{n-3} \rightarrow \dots \rightarrow C_0 \rightarrow C_{-1} \rightarrow 0$$

same as $2^{[n]}$

$$\tilde{H}_k(2^{[n]}/[n]) \cong \tilde{H}_k(2^{[n]}) \text{ for } k \leq n-3$$

$$\dim \ker \partial_{n-2} = 1 \quad \text{im } \partial_{n-1} = 0$$

$$\tilde{H}_{n-2}(2^{[n]}/[n]) \cong \mathbb{F} \quad \tilde{H}_k(2^{[n]}/[n]) = 0 \text{ for } k \leq n-3$$

Prop: $H_k(\Delta) \cong \tilde{H}_k(\Delta)$ for all $k \geq 1$.

$H_0(\Delta) \cong \mathbb{F}^{c(\Delta)}$ where $c(\Delta)$ is the number of connected components of Δ

$\tilde{H}_0(\Delta) \cong \mathbb{F}^{c(\Delta)-1}$ if $\Delta \neq \{\emptyset\}$.

$H_{-1}(\Delta) \cong \tilde{H}_{-1}(\Delta) \cong 0$ if $\Delta \neq \{\emptyset\}$

$\tilde{H}_k(\{\emptyset\}) \cong \begin{cases} \mathbb{F} & \text{if } k = -1 \\ 0 & \text{otherwise.} \end{cases}$

Theorem: If Δ and Δ' are simplicial complexes with $|\Delta| \approx |\Delta'|$, then $H_\bullet(\Delta) \cong H_\bullet(\Delta')$.

→ true for all k

$$|2^{[n]}| \approx B^{n-1} \leftarrow (n-1)\text{-dim ball}$$

$$\tilde{H}_\bullet(B^{n-1}) = 0$$

$$|2^{[n]}/[n]| \approx S^{n-2} \leftarrow (n-2)\text{-dim sphere}$$

$$\tilde{H}_k(S^{n-2}) = \begin{cases} \mathbb{F} & \text{if } k = n-2 \\ 0 & \text{otherwise} \end{cases}$$

Proof: Use singular homology

Cellular homology

Thm: For any finite chain complex $\dots \rightarrow A_k \xrightarrow{d_k} A_{k-1} \rightarrow \dots$

$$\sum_k (-1)^k \dim A_k = \sum_k (-1)^k \dim H_k$$

Proof: $\dim H_k = \dim \ker d_k - \dim \operatorname{im} d_{k+1}$

$$\dim \operatorname{im} d_k = \dim A_k - \dim \ker d_k$$

$$\stackrel{\text{**}}{\dim A_k} = \dim \ker d_k + \dim \operatorname{im} d_k$$

Plug it in.

Cor: For any simplicial complex Δ ,

$$\sum_{k \geq 0} (-1)^k f_k(\Delta) = \sum_{k \geq 0} (-1)^k \dim H_k(\Delta)$$

"
Euler characteristic
 $\chi(\Delta)$

Cor: For any simplicial complex Δ ,

$$\sum_{k \geq -1} (-1)^k f_k(\Delta) = \sum_{k \geq -1} (-1)^k \dim \tilde{H}_k(\Delta)$$

"
reduced Euler characteristic
 $\tilde{\chi}(\Delta)$

$$\tilde{\chi}(B^{n-1}) = \tilde{\chi}(2^{[n]}) = 0$$

$$\tilde{\chi}(S^{n-2}) = \tilde{\chi}(2^{[n]} \setminus [n]) = (-1)^{n-2} = (-1)^n$$

$$\chi(S^{n-2}) = 1 + (-1)^n$$

Simplicial
manifolds

Recall that a **manifold** of dimension d is a topological space where every point has a neighborhood homeomorphic to $\mathbb{R}^d$. A **simplicial manifold** is a simplicial complex homeomorphic to a manifold.

Manifolds: S^{n-1} spheres

$\mathbb{R}P^n$ real projective space

$\mathbb{C}P^n$ complex projective space

$S^1 \times S^1$ torus

Given a simplicial complex Δ and a face $\sigma \in \Delta$, define the **link** of σ to be

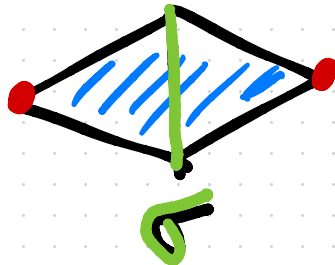
$$\text{link}_{\Delta} \sigma = \{ \tau \in \Delta : \tau \cup \sigma \in \Delta, \tau \cap \sigma = \emptyset \}$$

We define the **star** of σ to be

$$\text{star}_{\Delta} \sigma = \{ \tau \in \Delta : \tau \cup \sigma \in \Delta \}$$



star
link



Proposition: If Δ is a simplicial d -manifold, then for all $\sigma \in \Delta$,

$$H_*(IK_\Delta \sigma) \cong H_*(S^{d-1-\dim \sigma})$$

